Answer Key & Teacher Notes

Session 1 — Range and IQR · Session 2 — Comparing Variability

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Range and IQR

TALK PROMPTS

- Which number describes only the middle half of the data, and why?
- Why does one very large value change the range but not the IQR?

PARTNER TASK

Camp Hikes. One partner finds the five-number summary of the seven hikes; the other computes the range and the IQR from it. Swap roles with Bake Sale and compare which measure each partner trusts more.

WATCH FOR

- Computing the range (max − min) when the IQR is asked.
- Subtracting Q1 from the max, or the min from Q3, instead of Q3 − Q1.
- Assuming the IQR can never change when a value is changed or added; recompute the quartiles.

SESSION 2 · TEACHER NOTES

Comparing Variability

TALK PROMPTS

- Which measure convinced you that one class was more consistent, and why?
- When could two groups have the same range but very different IQRs?

PARTNER TASK

Phone Batteries. One partner finds the range and IQR for Phone 1; the other does Phone 2. Compare and agree on which phone is steadier, using both measures.

WATCH FOR

- Comparing the range of one group with the IQR of the other.
- Thinking a greater IQR always means a greater range.
- Calling the group with more values the more spread out group.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5c	1	1	1	3	2	2	3	3	2	1	1
S1 · Form B	6.SP.B.5c	1	1	1	3	2	2	3	3	2	1	1
S2 · Form A	6.SP.B.5c	2	1	3	3	3	3	2	3	2	1	1
S2 · Form B	6.SP.B.5c	2	1	3	3	3	3	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

range: The greatest value **minus** the least value.

interquartile range (IQR): Q3 minus Q1; the spread of the **middle half**.

extreme value: A value far from the others; it changes the **range** a lot.

- Range and IQR both measure **spread**, or variability.
- **For this seven-value example, increasing only the maximum leaves Q1 and Q3 unchanged. For a different data set, recompute both quartiles.**

Example: Order: 5, 8, 10, 12, 15, 18, 20. Range = 20 − 5 = **15**. Q1 = **8** and Q3 = **18**. Answer: IQR = 18 − 8 = **10** km

YOU TRY

1. Order: 4, 6, 7, 9, 11, 13, 15. a. 11 b. 13 − 6 = 7

2. 12 + 25 = 37

EXIT TICKET

1. Order: 3, 5, 7, 8, 10, 12, 14. a. 11 b. 12 − 5 = 7

2. 30 − 6 = 24; 22 − 10 = 12

3. **B** (Only the range changes)

PRACTICE FORM A

1. Order: 5, 8, 10, 12, 15, 18, 20. a. 20 − 5 = 15 b. 8 c. 18 d. 18 − 8 = 10

2. Order: 48, 51, 55, 59, 62, 64, 70. a. 70 − 48 = 22 b. 64 − 51 = 13

3. Order: 55, 61, 66, 70, 72, 79, 88. a. 88 − 55 = 33 b. 79 − 61 = 18

4. a. 21; 83 b. 15; 15 c. No. Only the range grows; Q1 and Q3 do not change, so the IQR stays 15.

5. a. 7 + 38 = 45 b. 32 − 15 = 17

6. a. 44 − 20 = 24 b. 40 − 24 = 16 c. 44 − 20 = 24 d. 42 − 22 = 20. Same range, but D's middle half is more spread out.

7. a. 62 − 11 = 51 b. 19 − 14 = 5 c. the IQR (the $62 wage stretches the range)

8. a. A: range 40, IQR 14; B: range 20, IQR 7 b. Bar B: its IQR is smaller. c. An unusual extreme can strongly affect the range. The IQR is often less sensitive; check whether Q1 or Q3 changes.

9. Tobias found the range (36 − 6 = 30). The IQR is Q3 − Q1 = 30 − 10 = 20.

10. **B** (greatest − least)

11. **B** (14)

Self-check word **EXTREME**

PRACTICE FORM B

1. Order: 6, 9, 11, 14, 17, 19, 22. a. 22 − 6 = 16 b. 9 c. 19 d. 19 − 9 = 10

2. Order: 52, 55, 58, 62, 66, 69, 74. a. 74 − 52 = 22 b. 69 − 55 = 14

3. Order: 50, 58, 62, 66, 68, 75, 84. a. 84 − 50 = 34 b. 75 − 58 = 17

4. a. 24; 80 b. 13; 13 c. No. Only the range grows; Q1 and Q3 do not change, so the IQR stays 13.

5. a. 9 + 41 = 50 b. 37 − 18 = 19

6. a. 40 − 10 = 30 b. 35 − 15 = 20 c. 40 − 10 = 30 d. 39 − 11 = 28. Same range, but D's middle half is more spread out.

7. a. 70 − 12 = 58 b. 21 − 15 = 6 c. the IQR (the $70 wage stretches the range)

8. a. A: range 30, IQR 11; B: range 46, IQR 22 b. Bar A: its IQR is smaller. c. An unusual extreme can strongly affect the range. The IQR is often less sensitive; check whether Q1 or Q3 changes.

9. Marisol found the range (40 − 5 = 35). The IQR is Q3 − Q1 = 31 − 9 = 22.

10. **B** (middle half)

11. **A** (17)

Self-check word **MEASURE**

GUIDED NOTES

consistent: Values are close together; the spread is **small**.

measure of variability: A number that tells how **spread out** data are.

extreme value: A value far from the rest; it stretches the **range**.

- Compare the **same** measure for both groups. The smaller value means more consistent.
- When one value is far from the rest, trust the **IQR** more than the range.

Example: Ranges: 18 < 35, so R is **less** spread overall. IQRs: 12 < 25, so R's **middle half** is closer together. Answer: Class **R** is more consistent.

YOU TRY

1. a. 12 b. 6; Set Q

2. Set A (its range is much larger than its IQR)

EXIT TICKET

1. a. 12 b. 4 (Y is more consistent)

2. Team 1 (IQR 6 < 14)

3. **B** (IQR)

PRACTICE FORM A

1. Class R: range 88 − 70 = 18, Q1 73, Q3 85, IQR 12. Class S: range 95 − 60 = 35, Q1 65, Q3 90, IQR 25. More consistent: Class R

2. a. 50 − 30 = 20 b. 46 − 33 = 13 c. 45 − 34 = 11 d. 43 − 37 = 6. Machine 2 is steadier.

3. Not right. V's IQR is 20, W's IQR is 5: V's middle half is much more spread out.

4. Not right. The middle halves match, but T's range is 24 and U's range is 88, so U is much more spread out overall.

5. The IQR. The IQR summarizes the middle half and is often less sensitive to extremes than the range. Recompute quartiles because it is not immune to changes.

6. a. Set N b. Set M (its range is much larger than its IQR)

7. a. 26 − 14 = 12 b. 24 − 16 = 8 c. 24 − 17 = 7 d. 23 − 18 = 5. Phone 2 is more consistent.

8. a. 13 − 9 = 4 b. 18 − 3 = 15 c. Bus 1 (smaller IQR) d. Bus 2 is sometimes on time, but its late times vary a lot (up to 40 min). Bus 1 has less spread in the middle half of the observed delays; this does not guarantee an individual delay.

9. One measure does not decide the other. Example: Set A = 2, 4, 6, 10, 14, 16, 18 has IQR 12 and range 16, while Set B = 0, 8, 9, 10, 11, 12, 40 has IQR 4 and range 40. Check both measures.

10. **B** (closer together)

11. **A** (an extreme value)

PRACTICE FORM B

1. Team J: range 83 − 64 = 19, Q1 67, Q3 80, IQR 13. Team K: range 92 − 55 = 37, Q1 60, Q3 88, IQR 28. More consistent: Team J

2. a. 52 − 28 = 24 b. 48 − 31 = 17 c. 49 − 38 = 11 d. 47 − 40 = 7. Machine 2 is steadier.

3. Not right. V's IQR is 25, W's IQR is 6: V's middle half is much more spread out.

4. Not right. The middle halves match, but T's range is 22 and U's range is 83, so U is much more spread out overall.

5. The IQR. The IQR summarizes the middle half and is often less sensitive to extremes than the range. Recompute quartiles because it is not immune to changes.

6. a. Set M b. Set N (its range is much larger than its IQR)

7. a. 21 − 8 = 13 b. 19 − 10 = 9 c. 18 − 12 = 6 d. 17 − 13 = 4. Tablet 2 is more consistent.

8. a. 20 − 5 = 15 b. 13 − 10 = 3 c. Bus 2 (smaller IQR) d. Bus 1 is sometimes only 2 min late, but its times vary a lot (up to 35 min). Bus 2 has less spread in the middle half of the observed delays; this does not guarantee an individual delay.

9. One measure does not decide the other. Example: Set A = 2, 4, 6, 10, 14, 16, 18 has IQR 12 and range 16, while Set B = 0, 8, 9, 10, 11, 12, 40 has IQR 4 and range 40. Check both measures.

10. **B** (Set Y)

11. **B** (spread of the middle half)

Self-check word **COMPARE**

Self-check word **VARYING**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · RANGE AND IQR

1. a. 21 − 6 = 15 b. 60 − 38 = 22 c. 31 − 18 = 13 d. 56 − 40 = 16 e. 90 − 68 = 22 f. 25 − 7 = 18

2. a. 30 − 6 = 24 pages b. 22 − 10 = 12 pages

3. Ordered: 28, 35, 41, 47, 52, 55, 60. a. 60 − 28 = 32 b. Median 47; Q1 = 35, Q3 = 55; IQR = 55 − 35 = 20

4. a. 14 + 29 = 43 b. 20 + 11 = 31

5. Half of the range is 32 ÷ 2 = 16, but the IQR is 20. The IQR depends on Q1 and Q3, not on the min and max.

SHEET 2 · COMPARING VARIABILITY

1. a. 12 b. 6 (Set Q is more consistent)

2. Player 1: range 16 − 8 = 8, IQR 14 − 10 = 4. Player 2: range 22 − 2 = 20, IQR 18 − 6 = 12. Player 1 is more consistent (both medians are 12).

3. Ranges: A: 80 − 70 = 10; B: 95 − 55 = 40. More consistent: Group A (smaller range; both medians are 75)

4. a. Route 1: 40 − 5 = 35; Route 2: 25 − 8 = 17 b. Route 1: 18 − 12 = 6; Route 2: 20 − 10 = 10

5. Kwame. Route 2's range is smaller (17 vs. 35), but Route 1's range is large only because of one extreme ride (40 min). Route 1's middle half is closer together (IQR 6 vs. 10), and the IQR is the better measure when there is an extreme value.

6. **A** (Class A)

7. **A** (6)

October 2026 conceptual additions — teacher key

Session 1 · Range and IQR

A: range 3, Q1 1.5, Q3 3.5, IQR 2. B: range 39, Q1 1.5, Q3 21.5, IQR 20. No; in these small sets the maximum participates in Q3. The seven-value examples have different quartile positions.

Extension: Q1 is 2 and Q3 is 6 in both seven-value sets. Their IQR is 4, although range increases from 6 to 69.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Variability

A: range 38 and IQR 4. B: range 20 and IQR 10. B has the smaller overall spread; A has the smaller middle-half spread. A claim must name the measure and purpose.

Extension: A has the narrower middle half and lower median, but its maximum is much larger. B could be preferred by someone concerned with the largest observed wait.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.